CLASSIFICATION

CLASSIFICATION TECHNIQUES – PART I

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Classification Techniques

The following concepts will be introduced:

- ✓ **Decision Tree**
- ✓ **node splitting**
- ✓ **Logistic regression**

Classification Techniques

A **Classification Technique (Classifier)** is a systematic approach to building **Classification Models** from a **Data Set**.

Classification Techniques can be grouped into:

- **Heuristic**; widely used like Decision Trees, Random Forests, Nearest Neighbor, …
- **Regression based**; use a parametric conditional probability, Logistic Regression
- **Separation**; partition the attributes' space, Support Vector Machines and Artificial Neural Networks
- **Probabilistic**; exploit Bayes formula and compute posterior probability to classify records, Naïve Bayes, Tree-Augmented Naïve Bayes, …

We provide a brief overview of some of this techniques even if a formal and detailed treatment of them is out of the scope of this course.

CLASSIFICATION TECHNIQUES

A **CLASSIFICATION TECHNIQUE (CLASSIFIER)** is a systematic approach to building **CLASSIFICATION MODELS** from a **DATA SET**.

CLASSIFICATION TECHNIQUES can be grouped into:

- **HEURISTIC**; widely used like DECISION TREES, RANDOM FORESTS, NEAREST NEIGHBOR, …

- **REGRESSION BASED**; use a parametric conditional probability, LOGISTIC REGRESSION

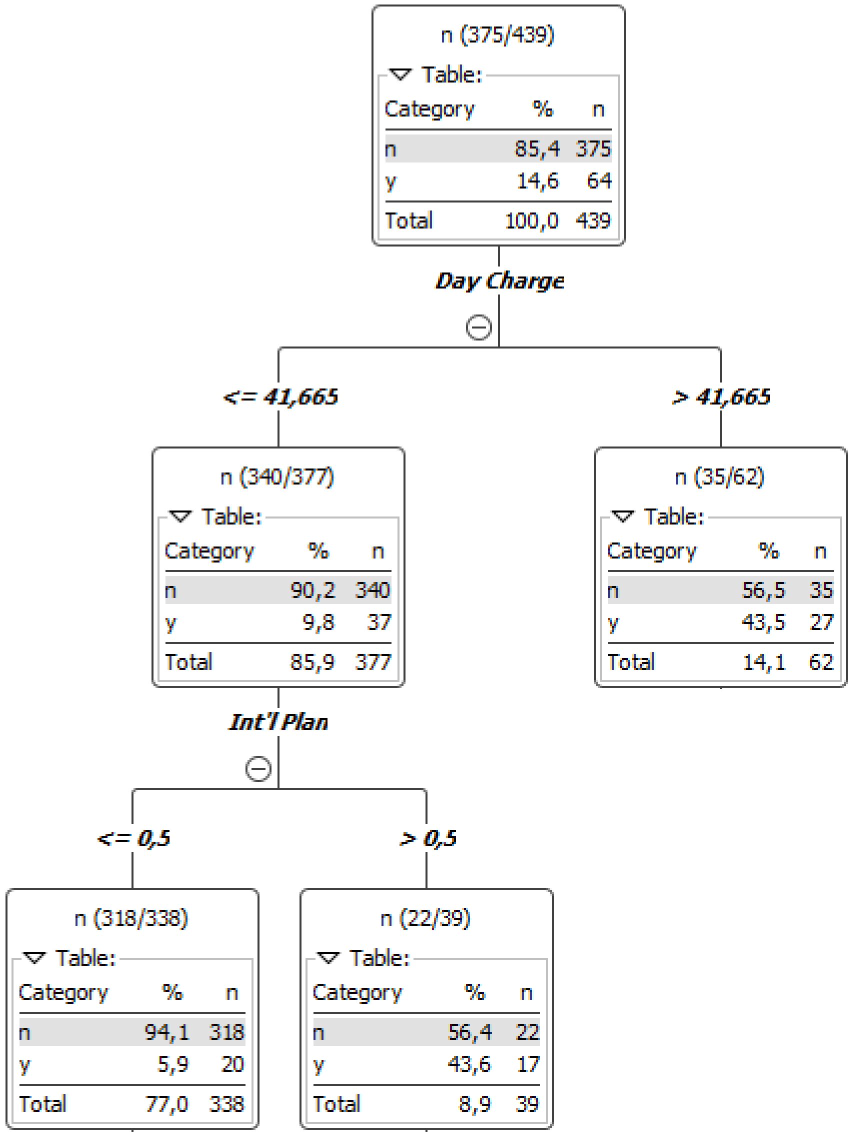
- **SEPARATION**; partition the attributes' space, SUPPORT VECTOR MACHINES and ARTIFICIAL NEURAL NETWORKS

- **PROBABILISTIC**; exploit Bayes formula and compute posterior probability to classify records, NAÏVE BAYES, TREE-AUGMENTED NAÏVE BAYES, …

We provide a brief overview of some of this techniques even if a formal and detailed treatment of them is out of the scope of this course.

Heuristic: Decision Trees

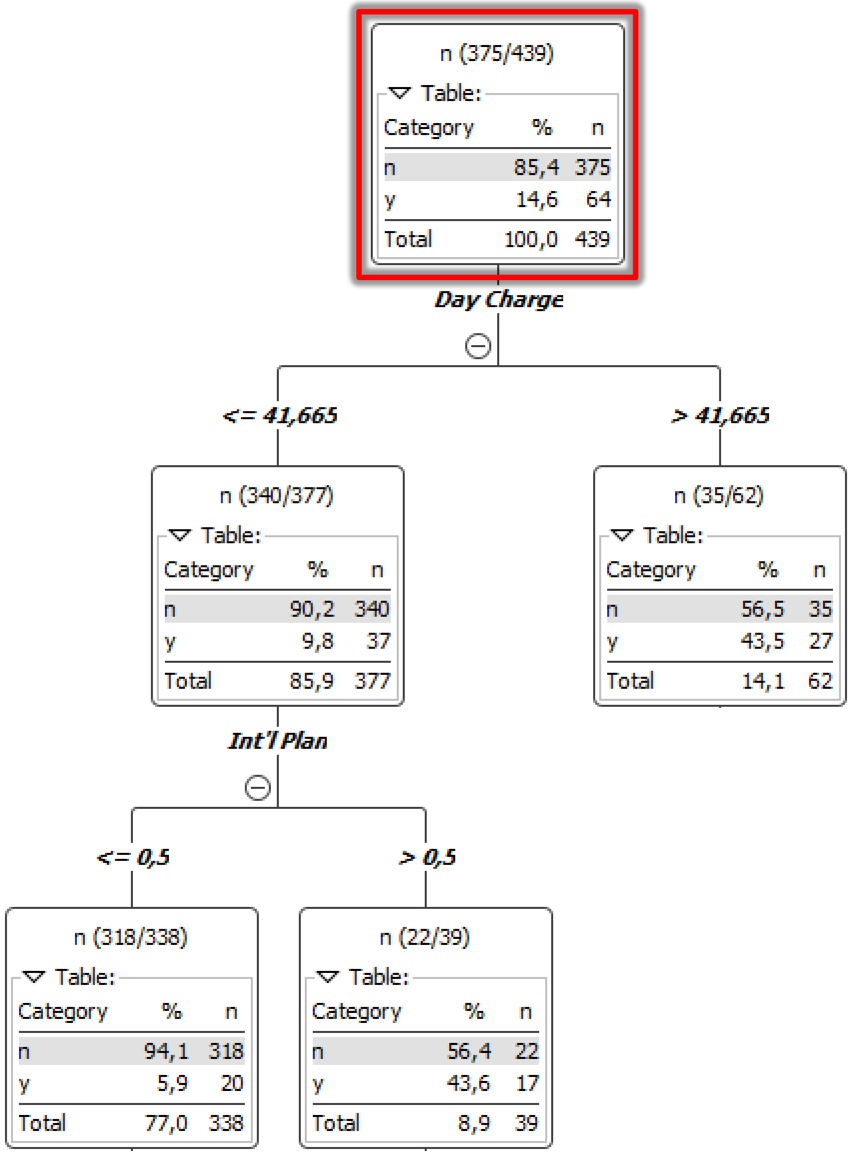
A **Decision Tree** consists of



HEURISTIC: DECISION TREES

A **DECISION TREE** consists of

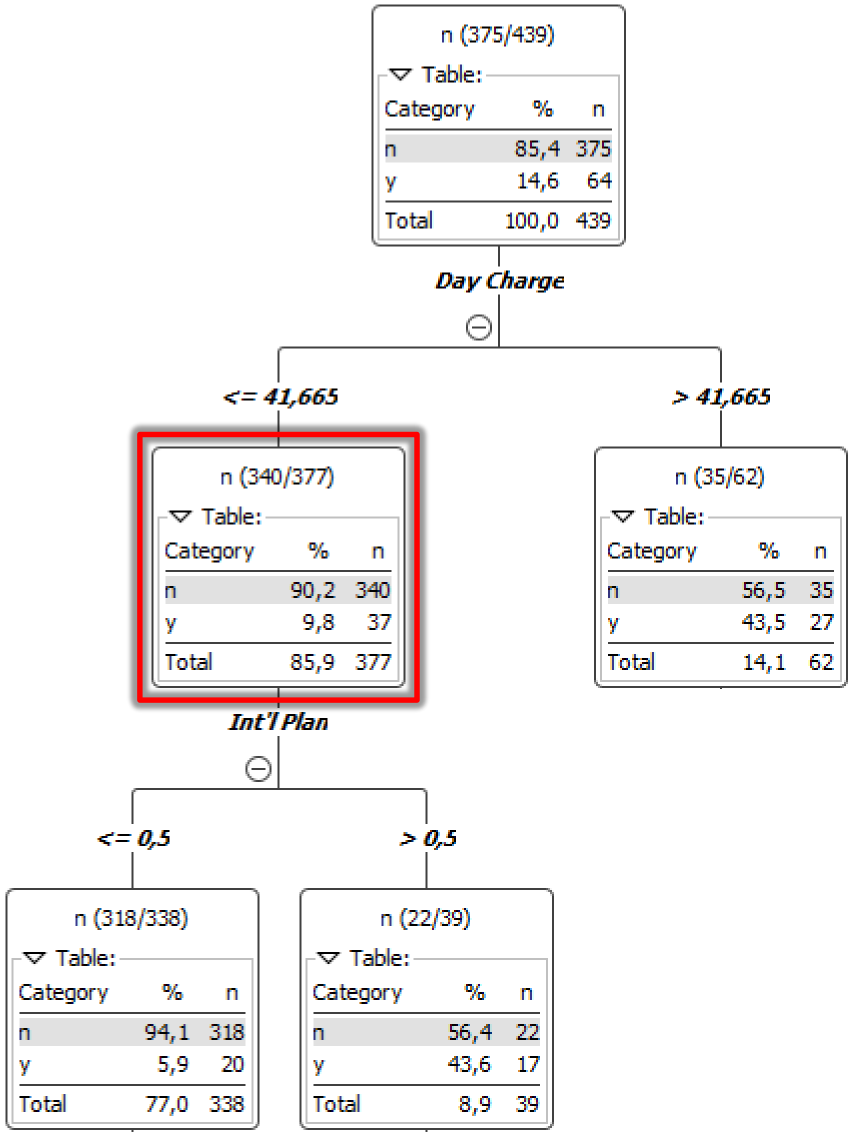
- **A ROOT NODE**; has no incoming edges and zero or more outgoing edges



Heuristic: Decision Trees

A **Decision Tree** consists of

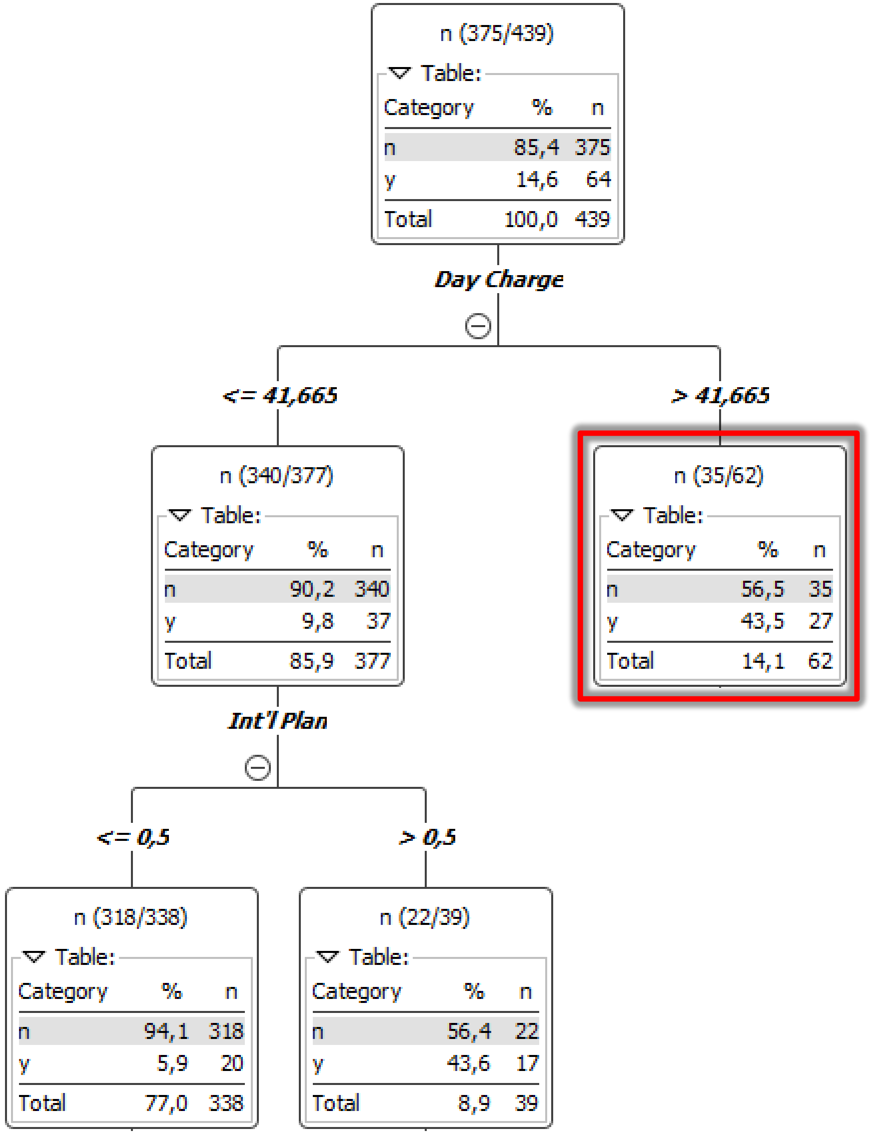
- **A Root Node**; has no incoming edges and zero or more outgoing edges
- **Internal Nodes**; each of which has exactly one incoming edge and two or more outgoing edges



HEURISTIC: DECISION TREES

A **DECISION TREE** consists of

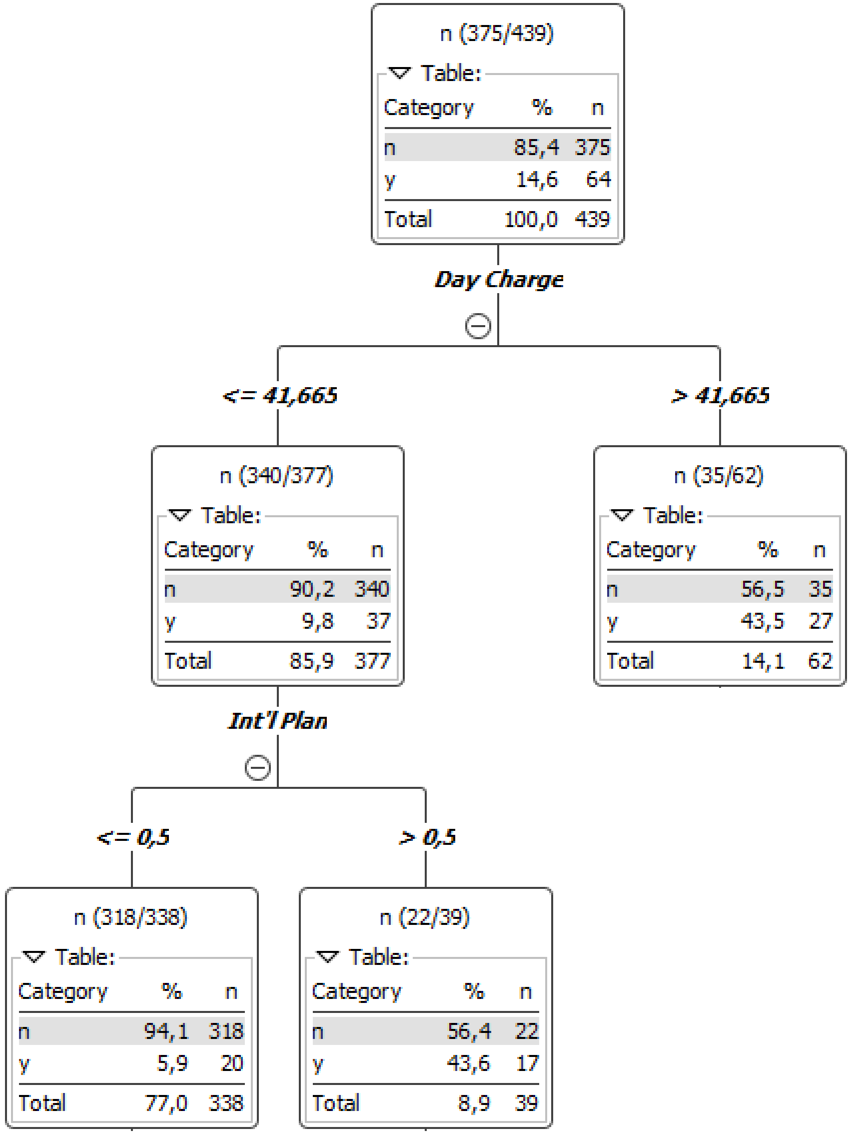
- **A ROOT NODE**; has no incoming edges and zero or more outgoing edges
- **INTERNAL NODES**; each of which has exactly one incoming edge and two or more outgoing edges
- **LEAF OR TERMINAL NODES**; each of which has exactly one incoming edge and no outgoing edges



HEURISTIC: DECISION TREES

2

Each record is classified from top to bottom



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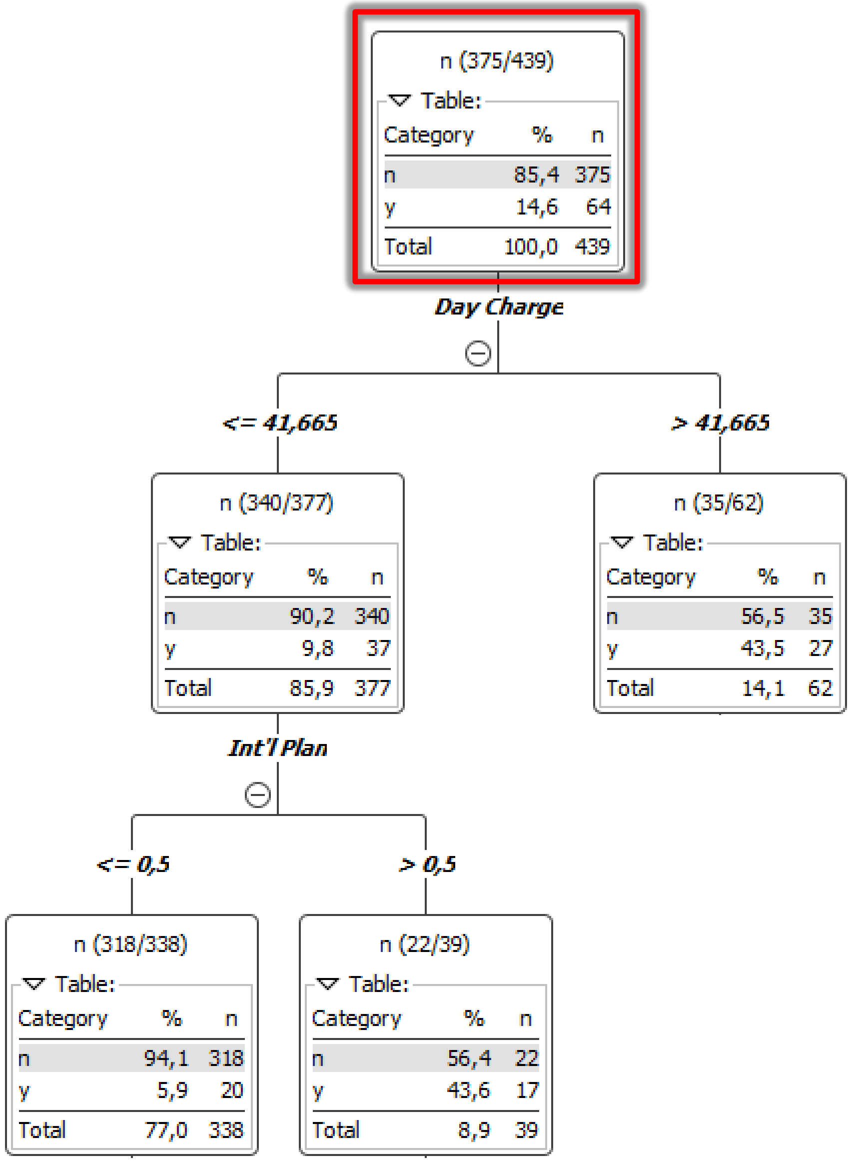
- **DAY CHARGE**; the following test is performed



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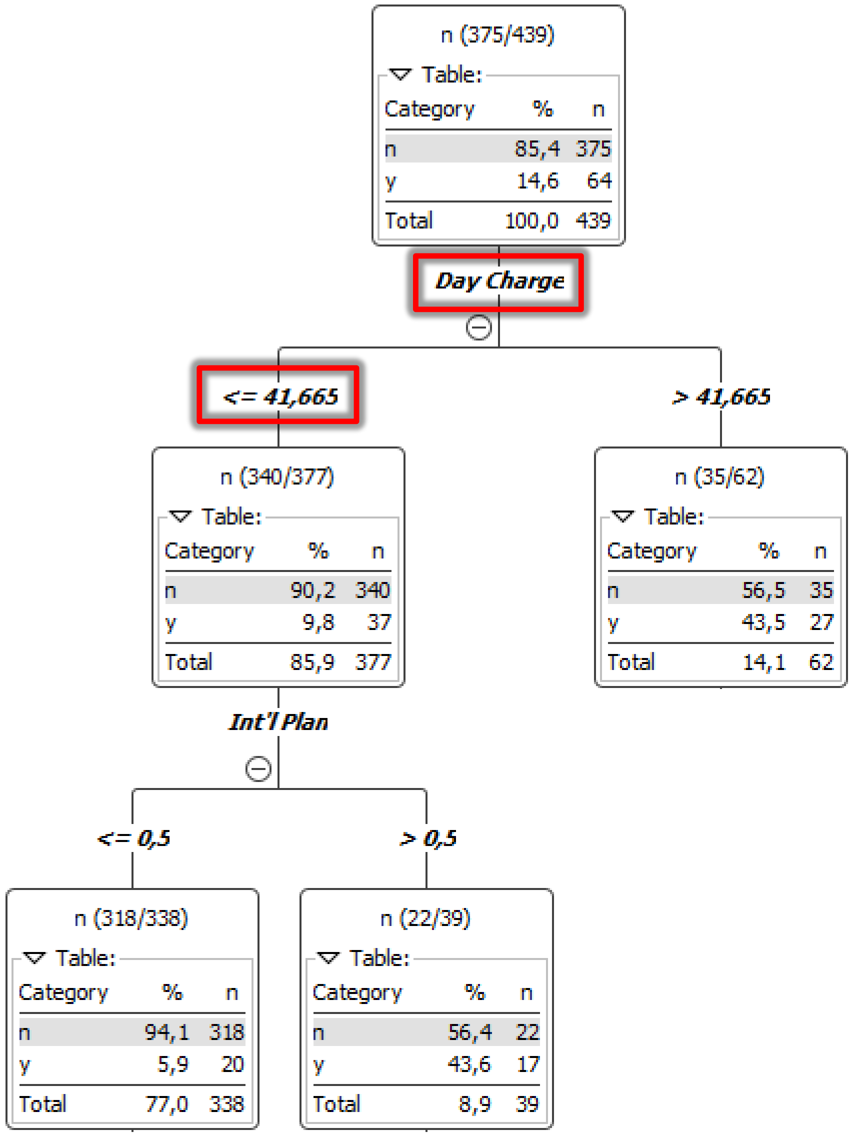
  IF DAY CHARGE ≤ 41.665 THEN



HEURISTIC: DECISION TREES

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  IF DAY CHARGE ≤ 41.665 THEN

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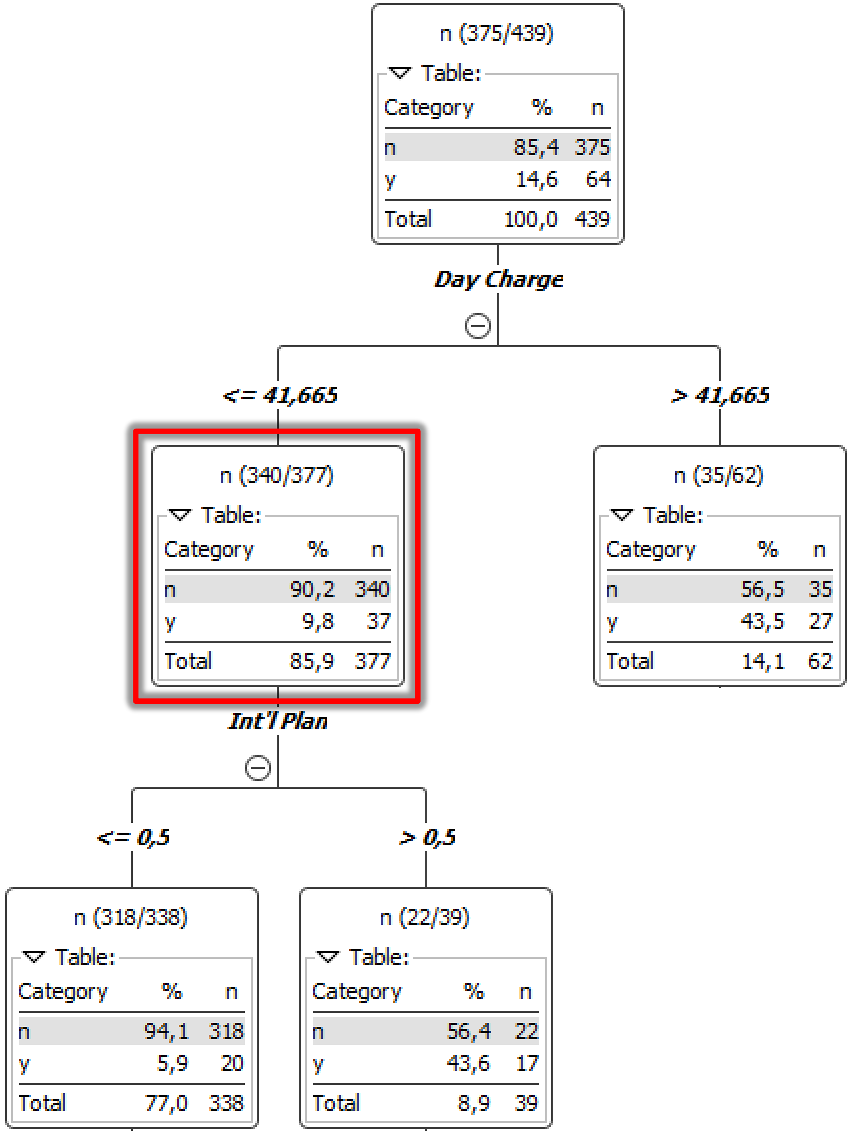

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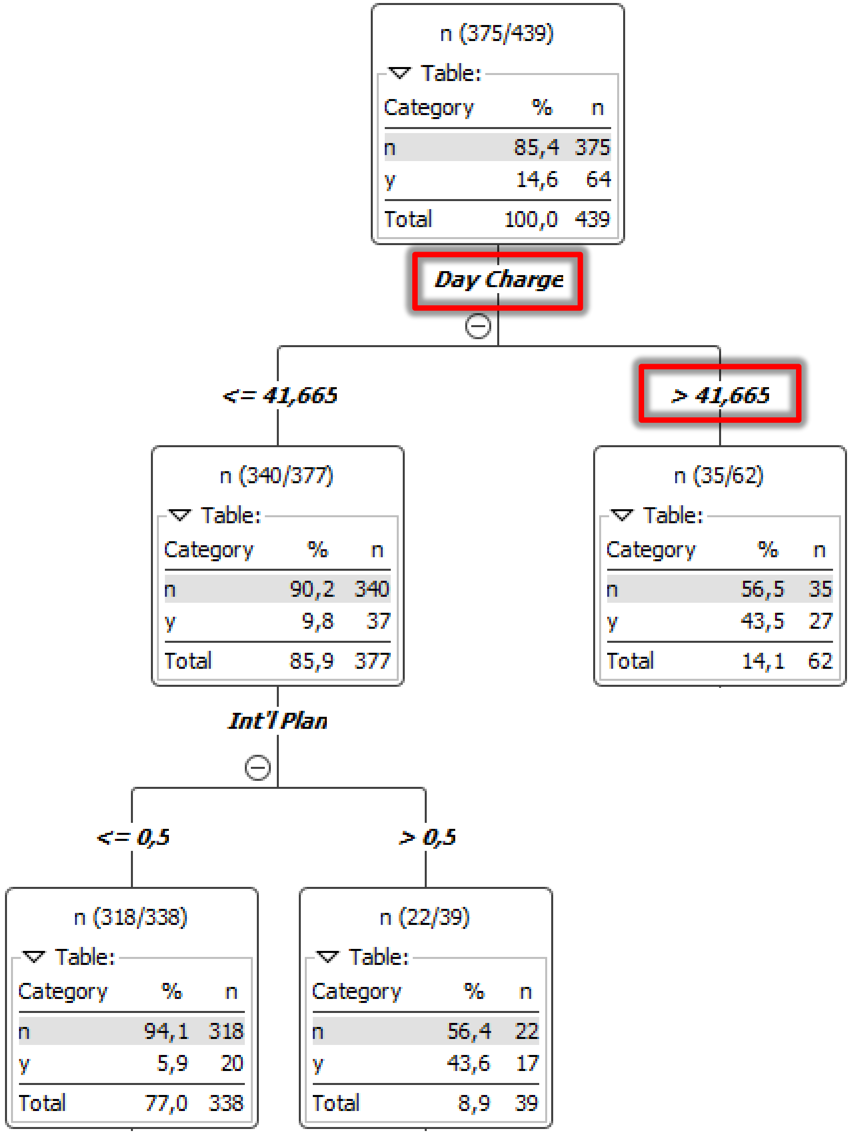

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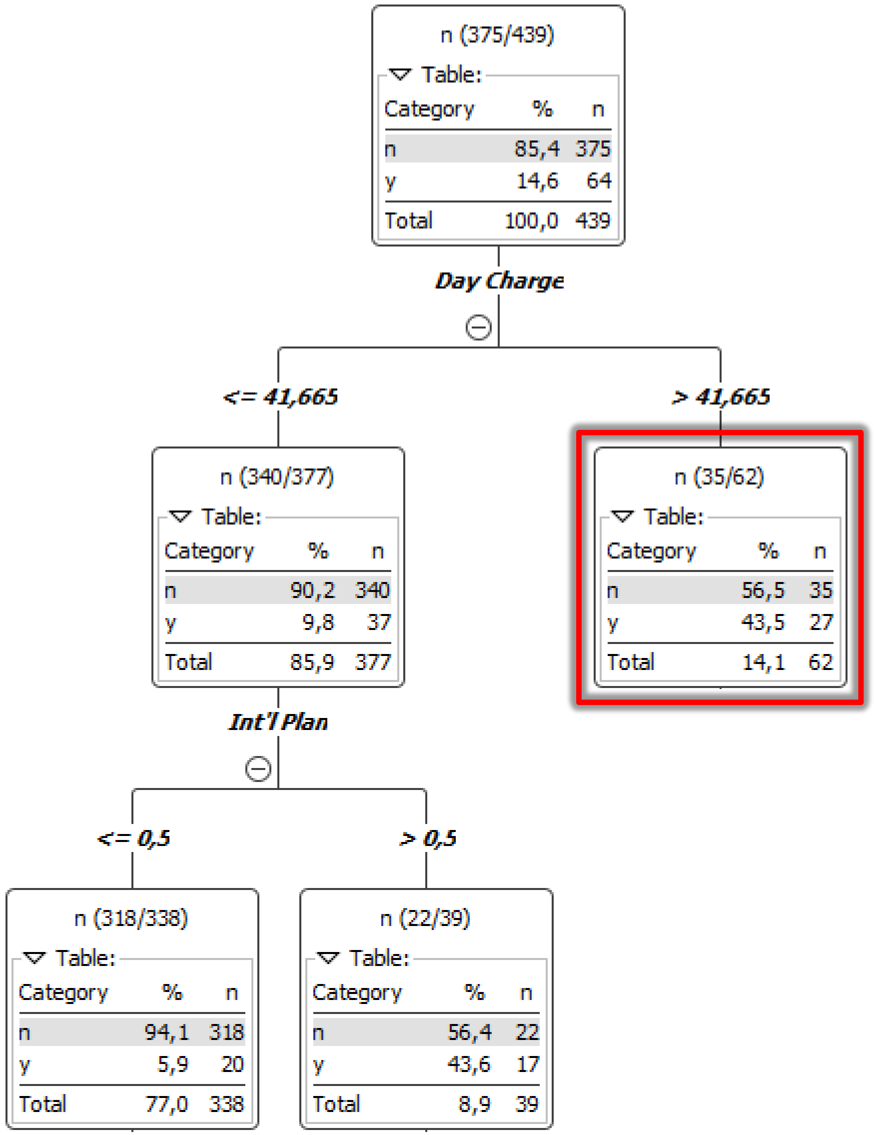

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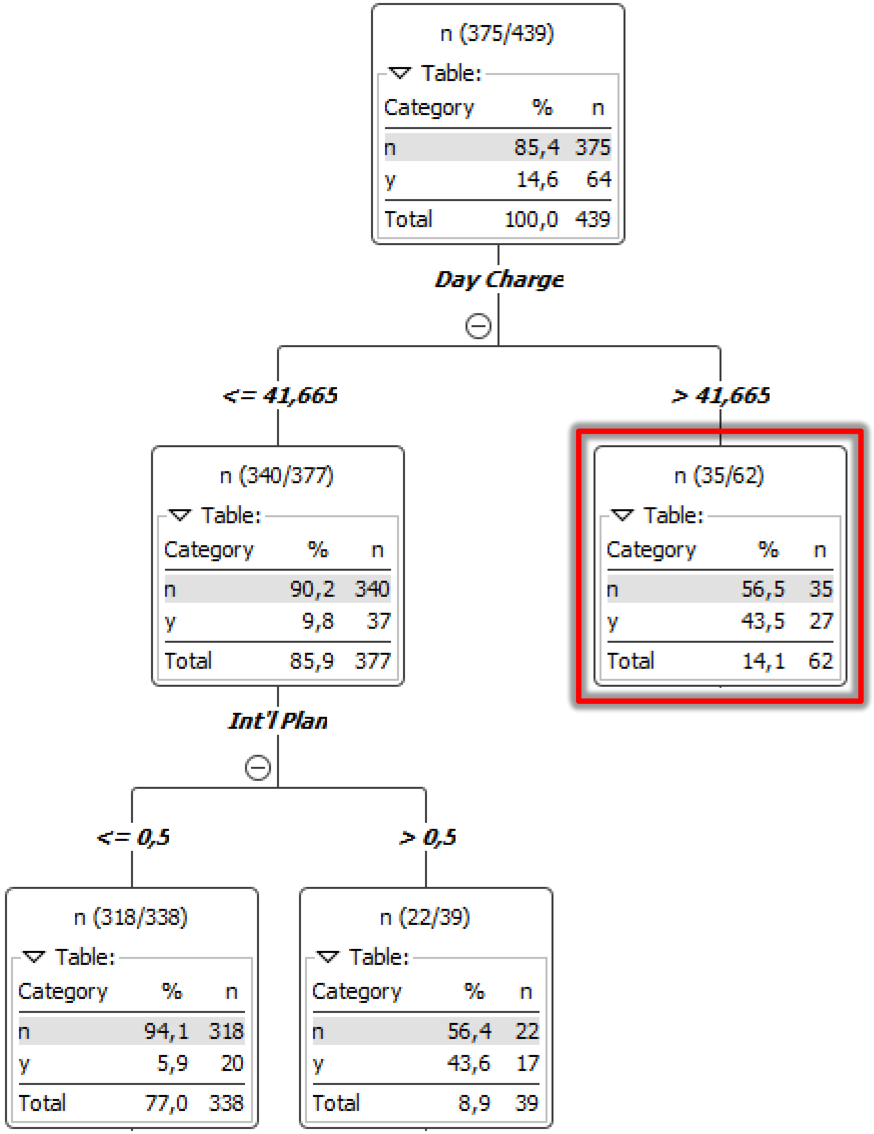
IF the node is a leaf node **THEN**

LABEL THE CURRENT RECORD WITH THE MOST FREQUENT CLASS ASSOCIATED WITH THE LEAF NODE



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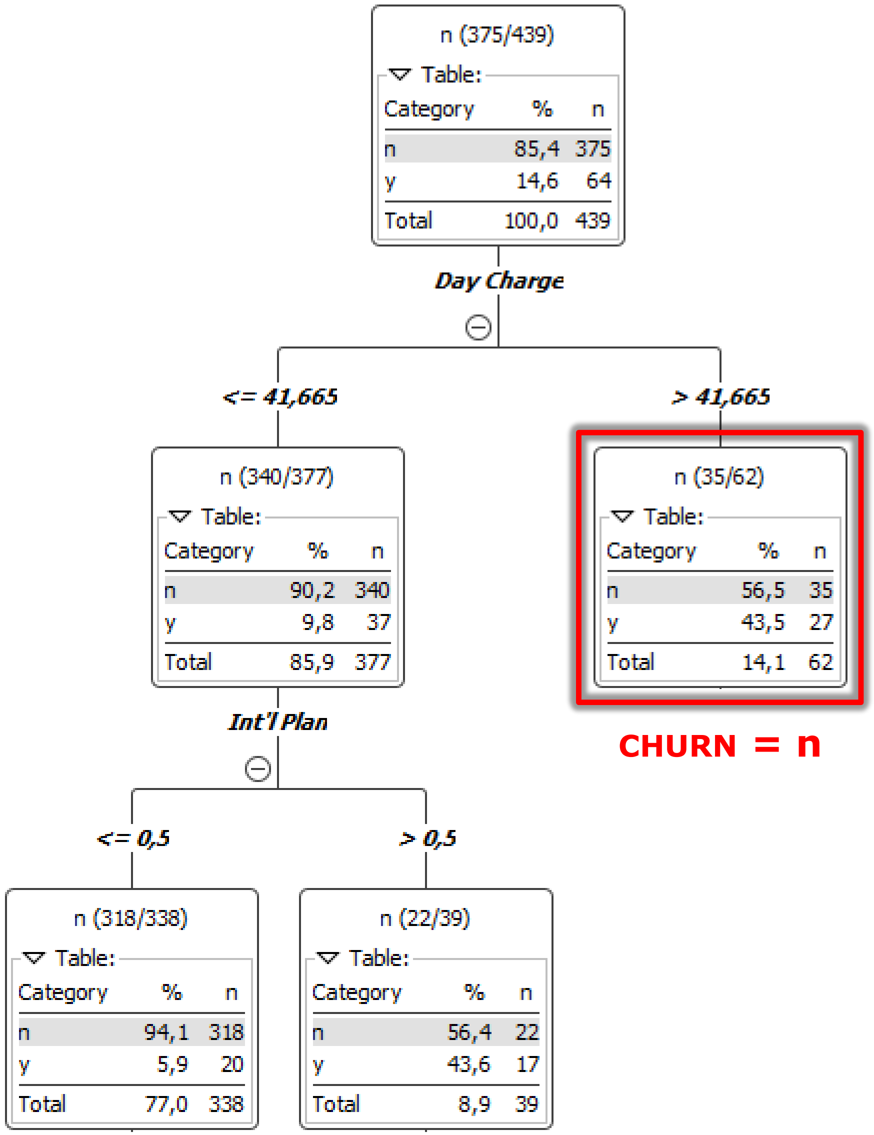

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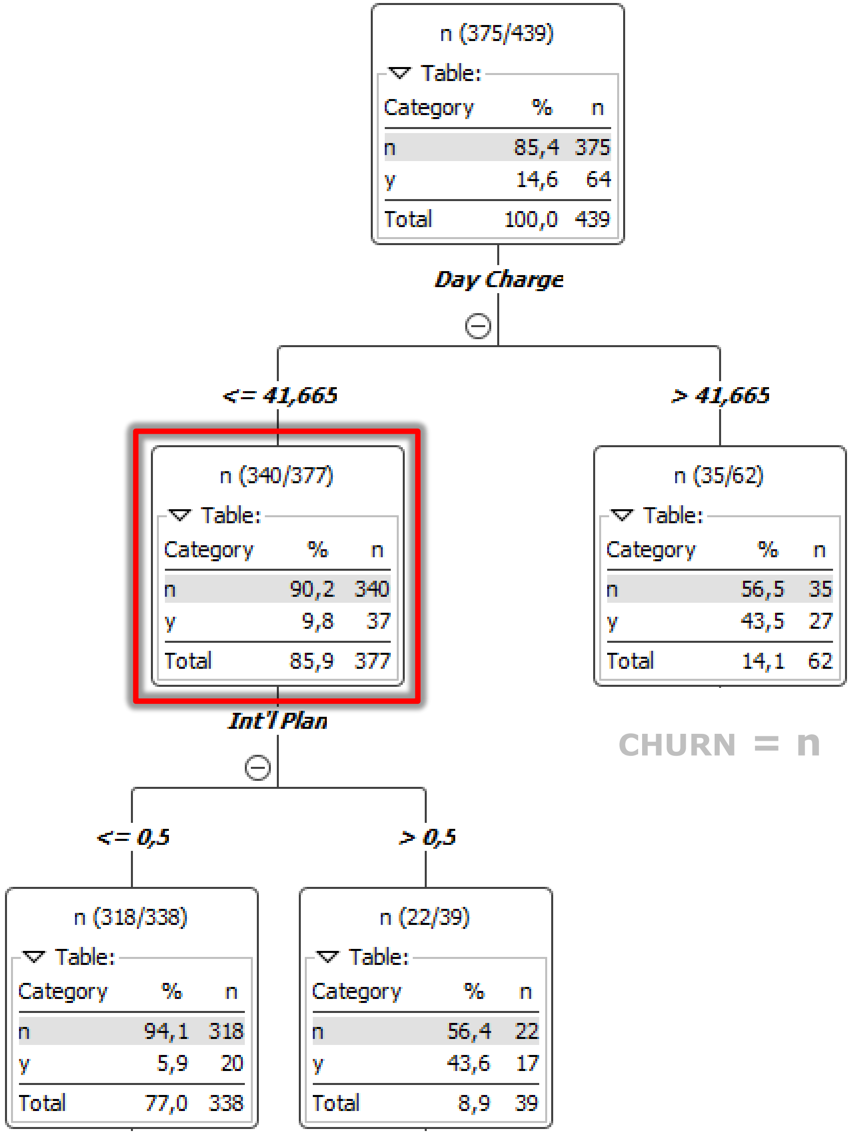
  Perform the test associated with the node



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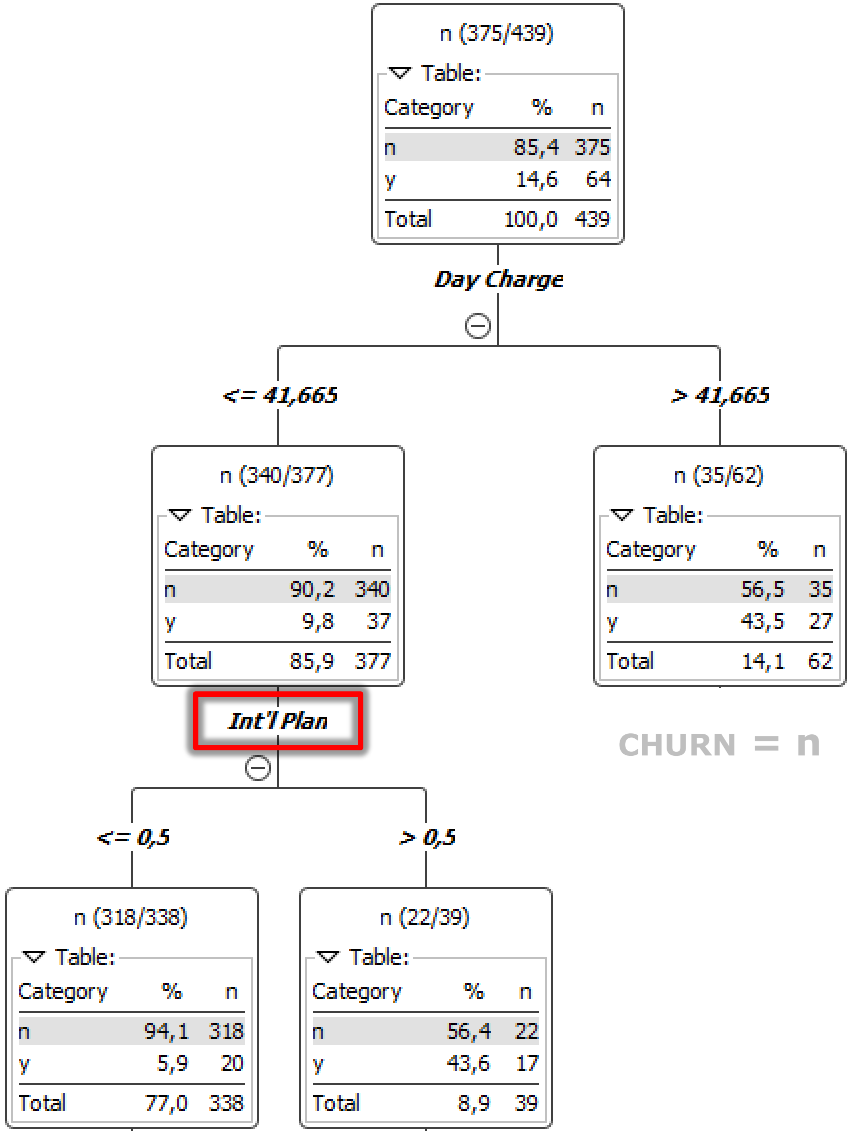
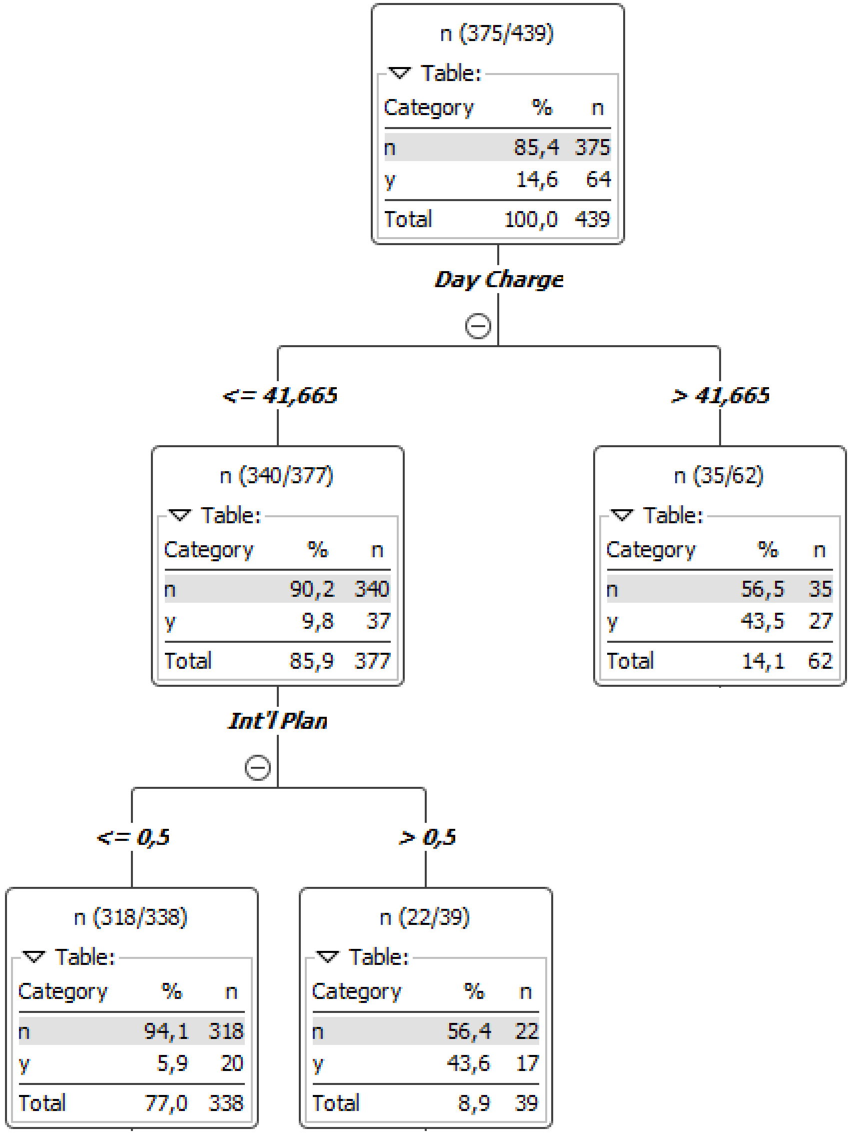


HEURISTIC: DECISION TREES

A Decision Tree can be used for CATEGORICAL NOMINAL AND ORDINAL ATTRIBUTES as well as for NUMERIC INTERVAL AND RATIO ATTRIBUTES.

Different measures can be used for selecting the **OPTIMAL NODE SPLITTING**

- **ENTROPY**
- **GINI IMPURITY INDEX**
- **CLASSIFICATION ERROR**



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Different measures can be used for selecting the **OPTIMAL NODE SPLITTING**

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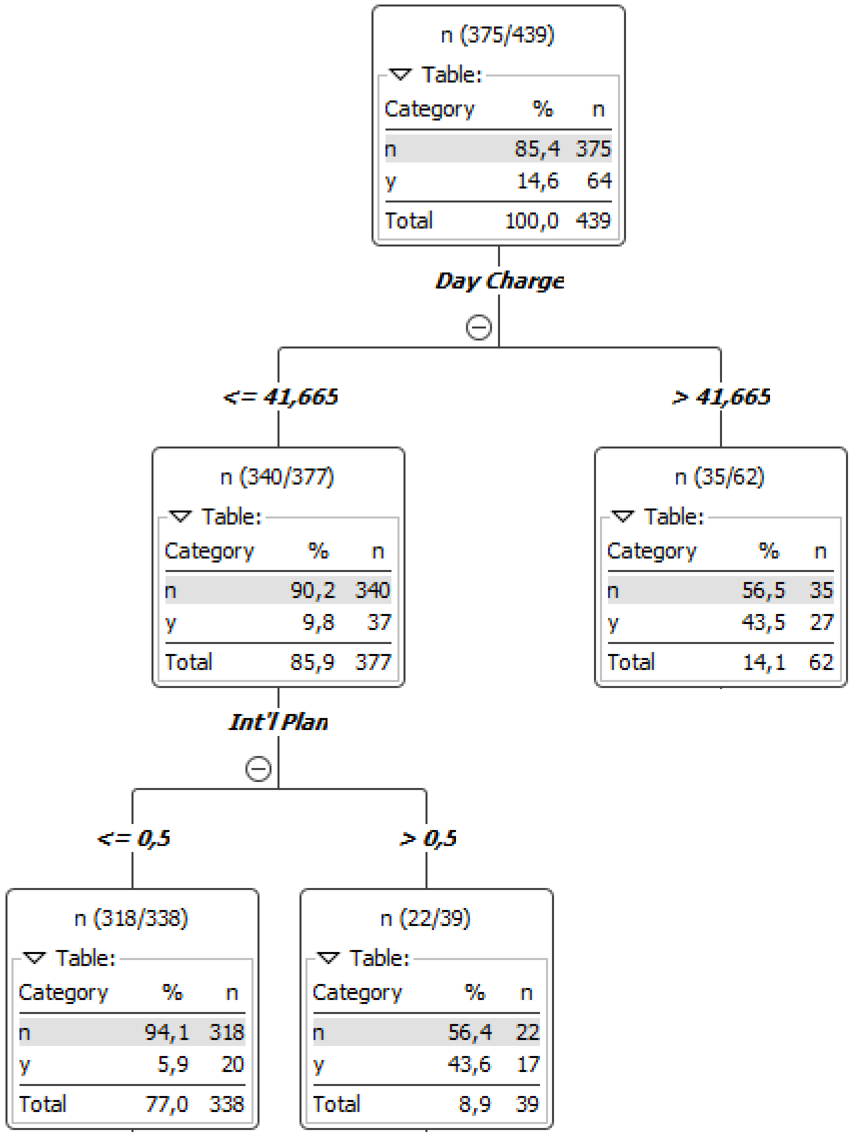
Node splitting depends on the type of the attribute

- **BINARY**
- **NOMINAL**
- **CONTINUOUS**



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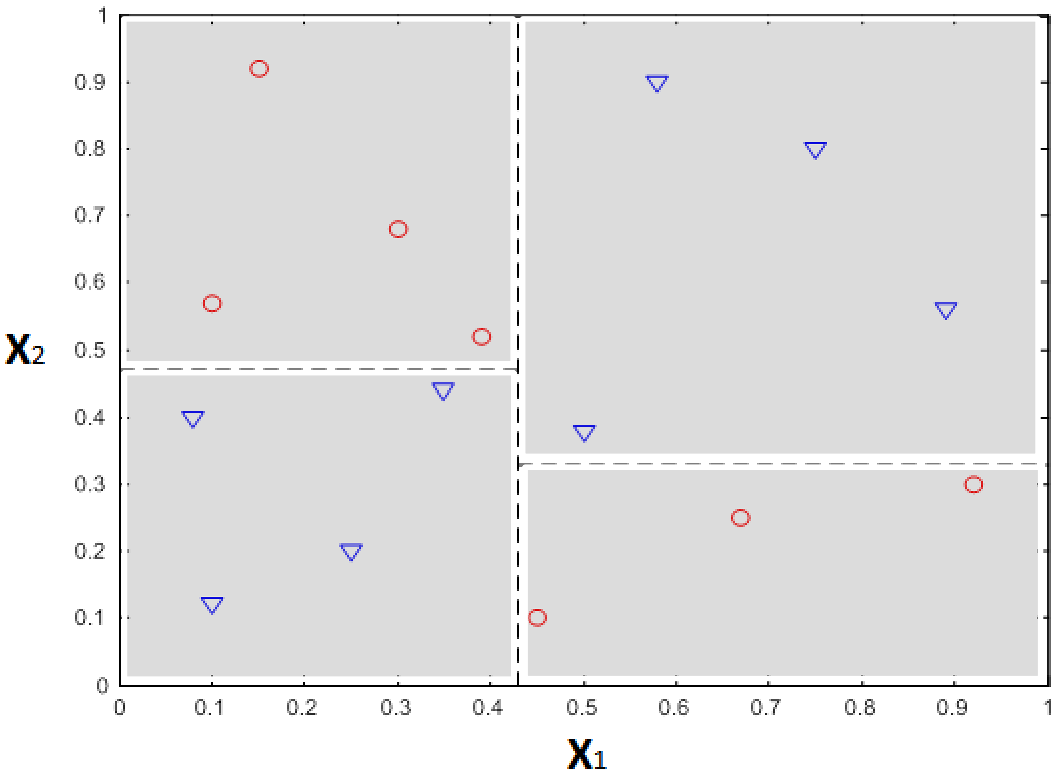
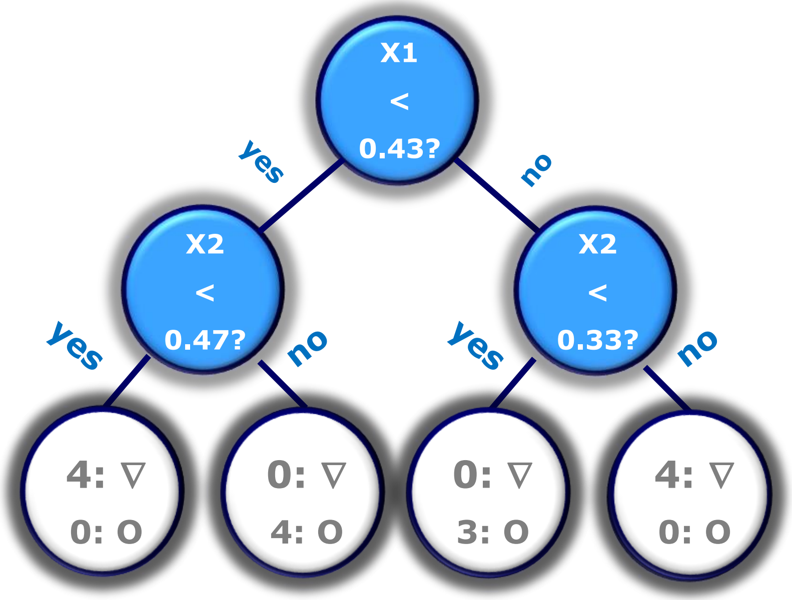
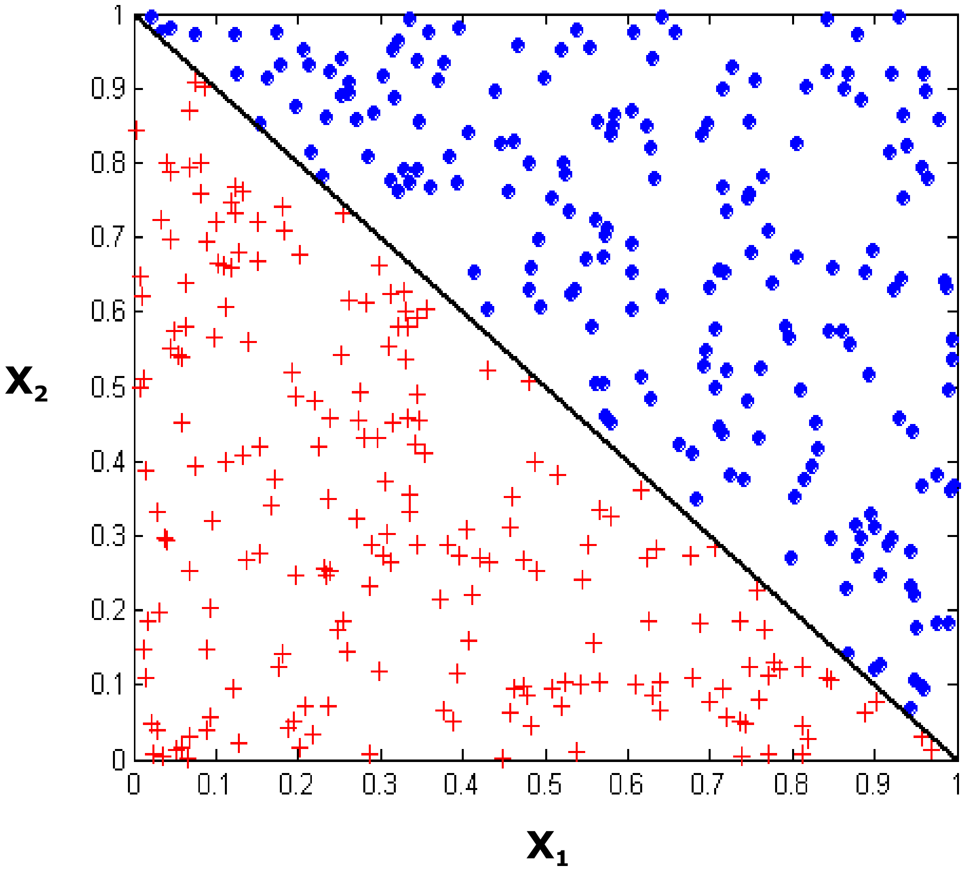


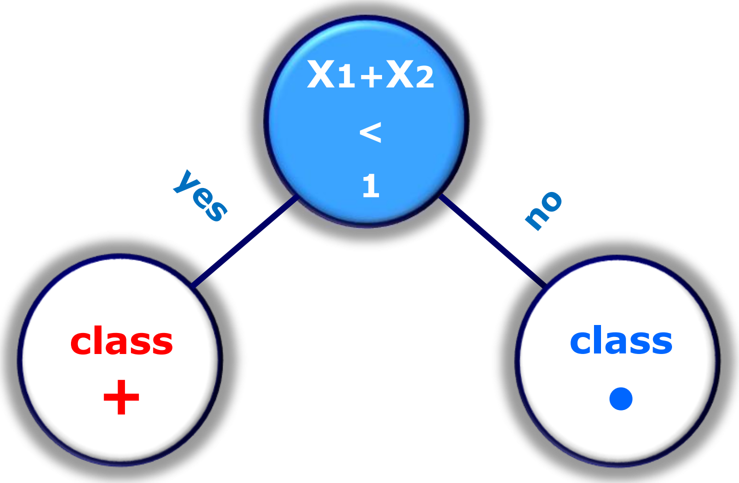


DECISION BOUNDARY, defines a change in classification.

Parallel to axes associated with attributes, the Decision Tree performs univariate tests to classify a record.

HEURISTIC: DECISION TREES





The testing condition can involve more than one attribute

Increased expressiveness

Computational demanding to find the optimal testing condition to perform

BINOMIAL LOGISTIC REGRESSION

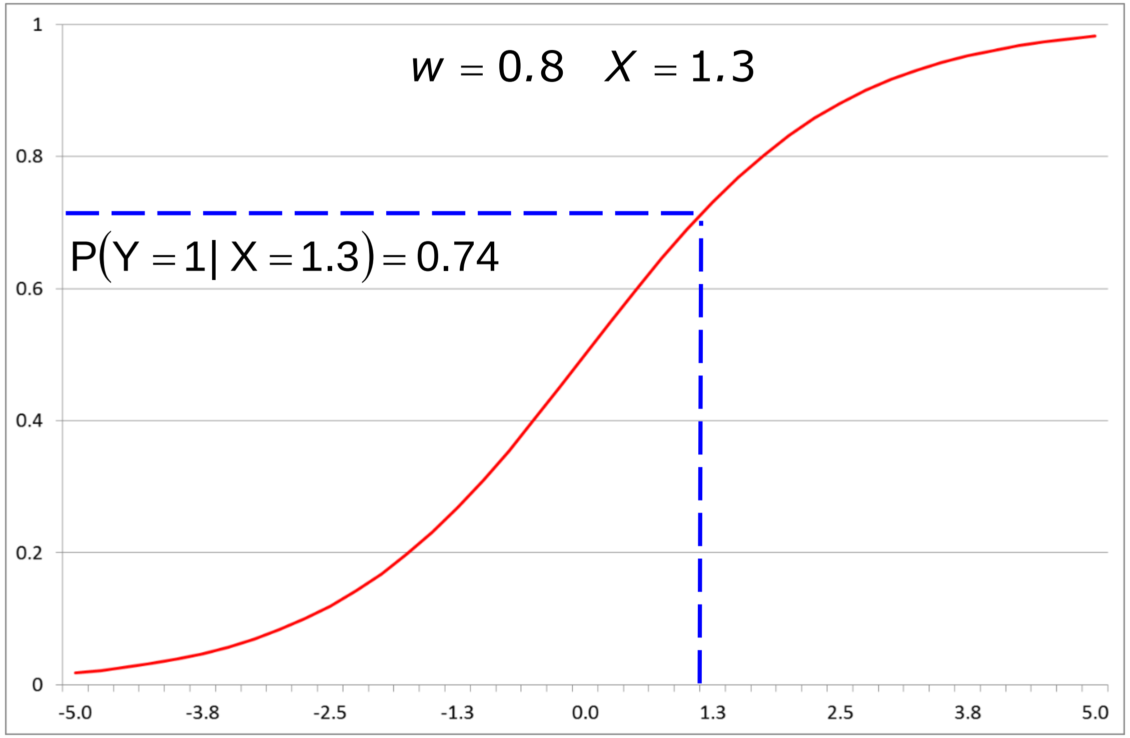
Assume the **CLASS ATTRIBUTE Y** takes value in {0,1}, then the binomial logistic regression classifer computes the posterior probability of the **CLASS ATTRIBUTE Y** given the value of the **INPUT OR EXPLICATIVE ATTRIBUTES X**.

In particular,

$$P(Y = 0 \mid \underline{X} = \underline{x}) = \frac{1}{1 + \exp(\underline{w} \cdot \underline{x})}$$

$$P(Y = 1 \mid \underline{X} = \underline{x}) = \frac{\exp(\underline{w} \cdot \underline{x})}{1 + \exp(\underline{w} \cdot \underline{x})}$$

where $\underline{w}$ is the **PARAMETERS VECTOR**.



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